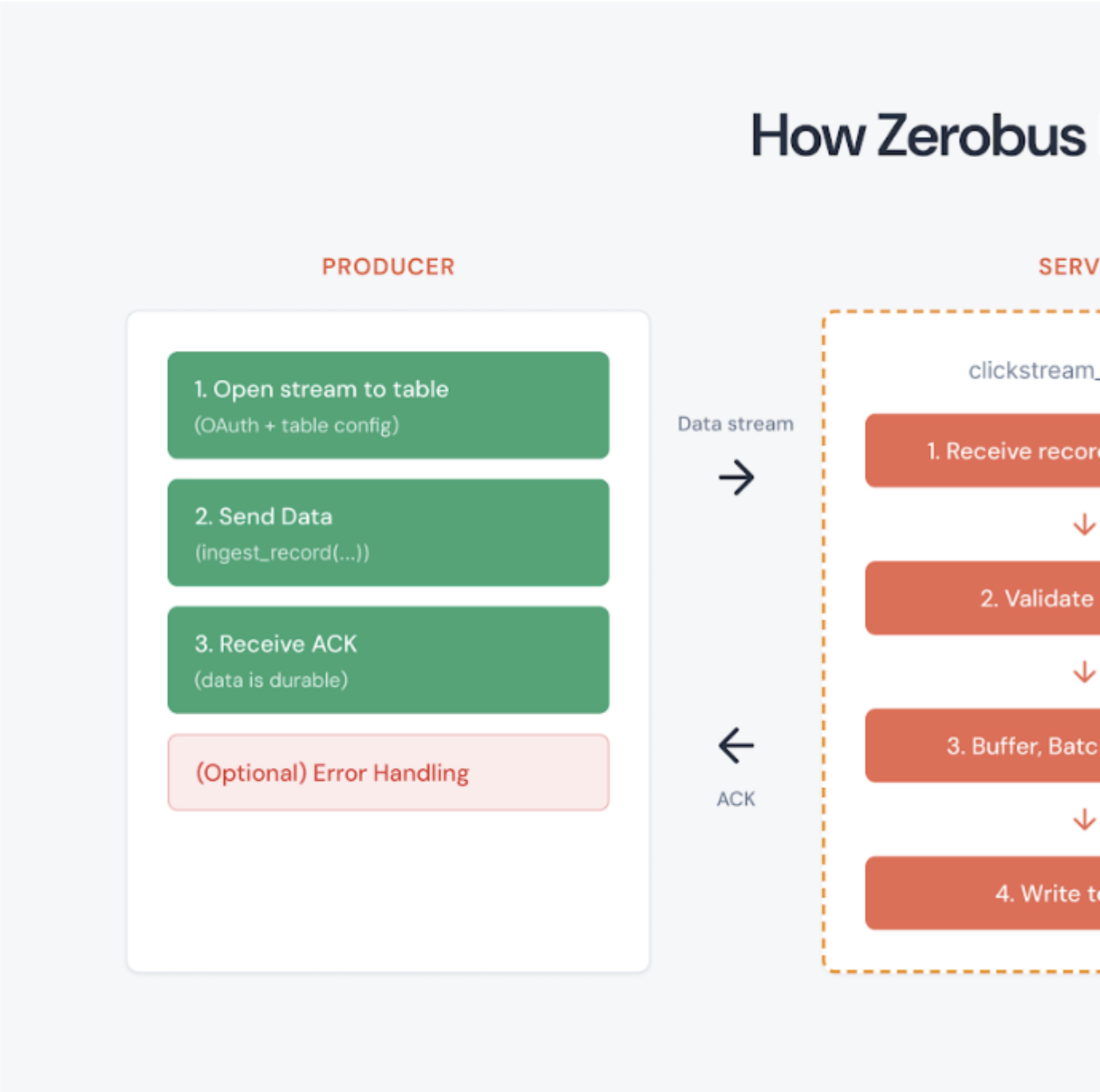
# zerobus-feeder

Configurable Python client that streams synthetic data into a Databricks Zerobus endpoint.
Useful for load-testing, demos, and smoke-testing Zerobus integrations.




Features:

- CLI flags, YAML config file, or interactive wizard (your last entries are remembered and prefilled).

- Random-data generator driven by a simple JSON schema (strings, ints, longs, floats, doubles, booleans, timestamps, dates, binary, choice lists, nullable columns).
- Optional helpers that create the target Delta table and the service principal (plus OAuth secret and grants) via a Databricks CLI profile.
- Live terminal dashboard with throughput, latency percentiles, and a sparkline of recent latencies.
- OAuth2 client-credentials authentication via the Zerobus Python SDK — no PAT on disk.

## Prerequisites

- Python 3.9+
- A Databricks workspace in a Zerobus-supported region (AWS, Azure, or GCP)
- Target table as a **managed** Delta table in Unity Catalog (the script can create it for you with `--create-table`)
- A service principal with OAuth2 client-credentials and `USE CATALOG`, `USE SCHEMA`, `MODIFY`, `SELECT` on the target table (the script can create and grant with `--create-sp` / `--create-table`)
- Databricks CLI configured (`~/.databrickscfg`) if you want to use `--create-sp` / `--create-table` or auto-prefill `workspace_url` / `workspace_id`

## Installation

### Option A: startup scripts (recommended)

The `run.sh` (Linux / macOS) and `run.bat` (Windows) wrappers create a local virtualenv on first launch, install dependencies, and then forward all arguments to `zerobus_feeder.py`:

```
# Linux / macOS
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
./run.sh
```

```
REM Windows
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
run.bat
```

Subsequent runs reuse the existing venv and skip reinstalling unless `requirements.txt` has changed.

### Option B: manual

```
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
python3 -m venv .venv
source .venv/bin/activate           # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

## Quick start (first run)

On the first invocation there is no remembered state, so the script offers a guided setup:

```
./run.sh                # or: run.bat   — or: python zerobus_feeder.py
```

The wizard will:

1. Let you pick a Databricks CLI profile from `~/.databrickscfg` and prefill `workspace_url` / `workspace_id` from it.
2. Optionally create a new service principal and OAuth secret.
3. Optionally create the target table (from the schema JSON) and apply grants.
4. Prompt for any remaining required fields and start streaming.

After the first run, values are written to `.zerobus_feeder_last.yaml` (already listed in `.gitignore`) and prefilled next time.

## Configuration precedence

| Source | Notes |
|---|---|
| `--config <yaml>` | If passed, **all other CLI flags and remembered values are ignored**. |
| CLI flags | Override remembered values. |
| `.zerobus_feeder_last.yaml` | Remembered values from the previous run. |
| Interactive prompts | Fill anything still missing (unless `--non-interactive`). |

## CLI reference

```
python zerobus_feeder.py [options]

  --config, -c FILE         Path to YAML config file (overrides all other params)
  --eps FLOAT               Target events (records) per second
  --schema-file FILE        JSON file describing the data structure
  --workspace-id DIGITS     Used to build the Zerobus endpoint
  --region CODE             e.g. us-west-2, eastus
  --cloud {aws,azure,gcp}   Cloud provider
  --table-name NAME         Full table name (catalog.schema.table)
  --client-id ID            Service principal client ID
  --client-secret SECRET    Service principal client secret
  --workspace-url URL       https://<host>; auto-filled from CLI profile if set
  --profile NAME            Databricks CLI profile (for --create-sp/--create-table)
  --warehouse-id ID         SQL warehouse used by --create-table
  --sp-display-name NAME    Display name for --create-sp
  --create-sp               Create an SP + OAuth secret (needs --profile)
  --create-table            Create the target table from the schema (needs --profile)
  --no-table-latency        Skip the periodic ingest-latency SQL query
  --interactive             Force prompts for every parameter
  --non-interactive         Never prompt; error on any missing required parameter
```

Mandatory parameters: `eps`, `schema_file`, `workspace_id`, `region`, `cloud`, `table_name`, `client_id`, `client_secret`, `workspace_url`. In `--non-interactive` mode the script exits with code 2 and lists any missing ones.

## YAML config

`sample_config.yaml` documents every parameter. When passed via `--config`, it is the sole source of configuration:

```
python zerobus_feeder.py --config sample_config.yaml
```

## Data structure JSON

`sample_schema.json` shows the supported types and per-column options. The schema drives both the randomly generated payloads and the `CREATE TABLE` issued by `--create-table`.

| Type | Delta type | Options |
|---|---|---|
| `string` | `STRING` | |

| Type | Delta type | Options |
|---|---|---|
| | | `length` or `min_length` / `max_length`, `prefix`, `alphabet`, `choices` |
| `int` | `INT` | `min`, `max`, `choices` |
| `long` | `BIGINT` | `min`, `max`, `choices` |
| `float` | `FLOAT` | `min`, `max`, `precision` |
| `double` | `DOUBLE` | `min`, `max`, `precision` |
| `boolean` | `BOOLEAN` | `true_probability` |
| `timestamp` | `TIMESTAMP` | `offset_seconds`, `jitter_seconds` (emitted as Unix microseconds) |
| `date` | `DATE` | `offset_days`, `range_days` (emitted as epoch days) |
| `binary` | `BINARY` | `length` (base64-encoded in JSON) |

Every column also accepts `nullable: true` + `null_probability: 0.0..1.0`.

## Creating the service principal and table

```
python zerobus_feeder.py \
  --profile DEFAULT \
  --schema-file sample_schema.json \
  --table-name main.default.zerobus_feeder_events \
  --cloud aws --region us-west-2 --workspace-id 1234567890123456 \
  --create-sp --create-table --eps 50
```

This will:

1. Create a service principal named `zerobus-feeder` (override with `--sp-display-name`), generate an OAuth secret, and store the credentials in `.zerobus_feeder_last.yaml`.
2. Prompt for (or auto-pick) a SQL warehouse, issue `CREATE TABLE IF NOT EXISTS <table>`, then grant `USE CATALOG` / `USE SCHEMA` / `MODIFY` / `SELECT` to the SP.
3. Start the stream at 50 EPS with the live dashboard.

If the workspace-level OAuth secret API isn't available, the script prints a link to the Databricks UI so you can generate the secret manually, then rerun with `--client-id` / `--client-secret`.

## Non-interactive / automated runs

After a first successful run, you can reproduce the stream from remembered state:

```
python zerobus_feeder.py --non-interactive
```

Or from a YAML file (ideal for CI / ephemeral VMs):

```
python zerobus_feeder.py --config /path/to/config.yaml
```

## Live dashboard

While running, the terminal shows:

- Target vs. actual EPS (events per second), total sent, errors, elapsed time
- Current / min / max / p50 / p95 / p99 SDK round-trip latency (ms)
- Unicode-block sparkline of recent per-record latencies
- End-to-end ingest latency (event_time → file_modification_time) over the last 15 minutes, sampled via a SQL query against the target table after every probe batch — disable with `--no-table-latency`
- Target table, Zerobus endpoint, workspace URL

```
guido.[redacted]  ~/code/zerobus-f
Connecting endpoint=7[redacted]39.zerobu
Workspace URL https://adb-7[redacted]39.
Table guido.zerobus.feed2
Authenticating as client_id=5f2aa102-67cf-4
✓ Stream opened. Starting ingestion...

Status:                          • mea
Sent:                            6,900
Elapsed:                         0:17:
Latency now:                     198.6
Latency min / max:               121.0
Target:                          guido
Table latency (15m):             avg 2

                                 Probe
   321.5 ms

   249.2 ms

   193.2 ms
```

Press `Ctrl+C` to flush and close the stream cleanly; a final summary is printed.

### Ingest-latency query

When a `profile` and `warehouse_id` are configured (or `warehouse_id` is selected at startup when missing), the feeder executes this query once per probe cycle on the same SQL warehouse:

```
SELECT
  count(*)                                                                                A
  round(avg(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time)
  min(unix_timestamp(_metadata.file_modification_time)   - unix_timestamp(event_time))  A
  max(unix_timestamp(_metadata.file_modification_time)   - unix_timestamp(event_time))  A
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(eve
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(eve
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(eve
FROM <catalog>.<schema>.<table>
WHERE event_time >= current_timestamp() - INTERVAL 15 MINUTES;
```

The query is bounded by a 15s warehouse-side timeout, so a cold warehouse cannot stall streaming. Pass `--no-table-latency` to skip it entirely.

## Authentication

Zerobus is authenticated via **OAuth2 client-credentials (M2M)** using a Databricks service principal. `client_id` and `client_secret` are passed to the Zerobus SDK, which handles token acquisition and refresh. The Databricks CLI profile is only consulted for the optional `--create-sp` / `--create-table` helpers and to prefill `workspace_url` / `workspace_id`.

## Files

| File | Purpose |
|---|---|
| `zerobus_feeder.py` | The script |
| `run.sh` | Startup wrapper for Linux / macOS (creates venv, installs deps, runs the script) |
| `run.bat` | Startup wrapper for Windows (same behaviour as `run.sh`) |
| `sample_schema.json` | Example data-structure definition |

| File | Purpose |
| --- | --- |
| `sample_config.yaml` | Example YAML config |
| `requirements.txt` | Python dependencies |
| `.gitignore` | Excludes the last-values file, log file, venvs, caches |
| `.zerobus_feeder_last.yaml` | Auto-generated, local-only, stores last-used values **including the client secret** — never commit this |
| `zerobus_feeder.log` | Auto-generated, local-only, detailed per-session log (config resolution, SP/table creation, DDL, periodic latency snapshots, errors). Appended across runs. Never commit this |

## Troubleshooting

- **`Authentication failed` / `error 4024`**: the SP is missing explicit table-level `MODIFY` / `SELECT`. Schema-level inherited grants aren't sufficient for the Zerobus `authorization_details` flow.
- **`Connection refused` / DNS errors**: verify `cloud` and `region` match the workspace; the endpoint is built as `<workspace_id>.zerobus.<region>.<cloud-suffix>`.
- **Actual EPS well below target**: latency dominates at the single-stream throughput limit (100 MB/s, 15 k rows/s). Run multiple instances or lower the per-record payload.
- **First-run wizard doesn't appear**: it triggers only when `.zerobus_feeder_last.yaml` is absent and `--config` / `--non-interactive` aren't set. Delete the file to reset.

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